

Dispersed awards, persistent ties: evidence from Türkiye's public IT procurement

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Abstract

Aggregate award concentration and repeated buyer–supplier relationships describe different dimensions of public procurement. We examine 9,991 information-technology contracts retrieved from Türkiye's e-procurement platform between October 2010 and March 2026. Supplier concentration by contract count is low when awards are pooled over the observation period, although value-based concentration is higher in several annual product categories. Among contracts whose buyer had previously purchased in the same category, 45% go to a previous supplier, 1.7 times the rate under the most restrictive permutation benchmark. The excess persists under coarser classifications, category-free buyer histories, and later evaluation windows. Purchases with similar earlier titles account for most of the excess, but this title-based indicator does not establish contractual renewal. Conditional-logit estimates show large associations with both past and future buyer–supplier ties. Simulations illustrate how these patterns depend on history, persistent match differences, and initialization assumptions; they do not identify a causal switching-cost effect or decompose persistence into mechanisms. Incumbent wins coincide with more single bidding and, in a small estimated-cost sample, smaller but statistically imprecise discounts. Buyer-level persistence therefore complements concentration measures as a descriptive monitoring tool. Keyword coverage, AI-only classification checks, and unobserved bidder participation limit conclusions about market power and welfare.

Keywords: Public procurement, Incumbency, Buyer–supplier relationships, Permutation benchmarks, Market concentration, Türkiye

JEL Classification: H57 , L14 , L86 , D22

1 Introduction

Concentration and repeated buyer–supplier relationships describe different dimensions of procurement competition. The Herfindahl–Hirschman index (HHI) and concentration ratios summarize supplier shares within a defined market and time interval (Hirschman, 1964; Rhoades, 1993; U.S. Department of Justice and Federal Trade Commission, 1992, 2023). They do not describe which buyers return to which suppliers. Installed systems can make switching costly in public IT, allowing persistent relationships even where aggregate awards are dispersed (Borenstein, MacKie-Mason, & Netz, 2000; Farrell & Klemperer, 2007; Farrell & Shapiro, 1988; Greenstein, 1993; Klemperer, 1987; Shapiro, 1995). Yet neither low concentration in broad title categories nor repeated awards establishes the competitive conditions facing a particular buyer.

Persistence, however, is not the same as lock-in, by which we mean persistence caused by switching costs. A buyer may return to a supplier because switching is costly (state dependence), or because that supplier simply suits its needs, location, or systems, or because the two have a relationship that the buyer values or that favours the supplier (stable heterogeneity). Separating these explanations is a classic problem (Dubé, Hitsch, & Rossi, 2010; Heckman, 1981; Heckman & Borjas, 1980; Keane, 1997), and it matters for policy. Switching costs call for remedies that lower them. Heterogeneity may be benign, as with specialization and good matches, or it may reflect favouritism, which calls for scrutiny of discretion.

This paper measures buyer–supplier persistence in a keyword-defined sample of Türkiye’s public IT procurement and examines what award histories can reveal about its interpretation. We reconstruct 9,991 contracts from EKAP (October 2010–March 2026), linking 2,814 firms to 2,890 buyers. Four findings organize the analysis.

- First, count-based HHIs pooled over the observation period are below 1,000 in every title-defined product category. Annual value concentration is higher in several categories, so the pooled result does not imply that relevant markets are uniformly unconcentrated.
- Second, persistence is large. Of the contracts awarded by a buyer that had bought in the same market before, 45.2% went to an incumbent. That is 1.7 times the rate under the most restrictive permutation benchmark, which keeps each buyer’s demand and each firm’s wins fixed within product category, year, and the buyer province and the supplier’s prior modal award province.
- Third, about a third of eligible contracts have a similar purchase 6–18 months earlier and account for most of the excess. This operational recurrence indicator describes repeated purchases, not legal contract renewal.
- Fourth, both previous suppliers and firms forming future ties with a buyer have large conditional associations with winning. Simulated benchmarks illustrate the sensitivity of these associations to model and history assumptions. They cannot identify the relative causal contributions of switching costs, learning, specialization, or preferential relationships.

Persistence is highest in health information systems and under Article 21(b), the negotiated procedure for urgent needs, and single bids are far more common among incumbent-won tenders. In a sample of tenders for which we collected estimated costs,

incumbent-won contracts have smaller estimated-cost discounts, but the difference is imprecise.

Our contribution is a set of complementary transaction-level diagnostics, with explicit limits on their interpretation. Earlier studies relate procurement concentration to network structure (I.G. Fountoukidis, Antoniou, & Varsakelis, 2023; Pliatsidis, 2024), while I. Fountoukidis, Daffi, Antoniou, and Varsakelis (2026a) directly predicts future buyer–supplier ties from historical relationships in European procurement. The latter is a close predecessor: examining persistent ties is not itself new. We complement that predictive approach by comparing observed incumbency with increasingly constrained assignment nulls, examining title-based recurrence, and testing sensitivity to category definitions and observed history. The framework of I. Fountoukidis, Daffi, Antoniou, and Varsakelis (2026b) likewise connects concentration and recurrence. Our simulations are diagnostic exercises rather than a causal decomposition. The Turkish IT application adds evidence to research on political connections and procurement (Gürakar, 2016; Gürakar & Bircan, 2019; Kutlay & Yıldırım, 2026), alongside a documented audit of the scope and valuation of keyword-retrieved data.

2 Background

2.1 Switching costs, heterogeneity, and incumbency

The HHI depends on product, geographic, and temporal boundaries. Our title categories organize comparable records but do not establish substitution-based relevant markets (European Commission, 2024; Rhoades, 1993). Switching-cost and aftermarket models explain how suppliers can possess advantages within installed systems despite modest aggregate shares (Borenstein et al., 2000; Farrell & Shapiro, 1988; Klemperer, 1987; Shapiro, 1995; Shapiro & Varian, 1999); the European Commission has raised this concern for public IT (European Commission, 2013). Recurrent procurement creates links between successive contests through learning, entry costs, and buyer switching costs (Iossa & Waterson, 2019). Contract duration itself trades off supplier reselection costs against remaining with an inefficient supplier (MacKay, 2022). Consequently, recurring titles motivate a separate descriptive comparison but do not isolate switching costs. Distinguishing state dependence from persistent differences requires assumptions about history and initial conditions (Dubé et al., 2010; Heckman, 1981; Heckman & Borjas, 1980; Keane, 1997; Wooldridge, 2005).

In procurement, incumbents can hold cost or information advantages that shape entry and bidding (De Silva, Dunne, & Kosmopoulou, 2003; Hyttinen, Lundberg, & Toivanen, 2018; Jofre-Bonet & Pesendorfer, 2003; Li & Zheng, 2009; Saini, 2012), and limited competition in government purchasing is common (Kang & Miller, 2022). Buyer discretion and the people who exercise it matter (Best, Hjort, & Szakonyi, 2023). More discretion raises the share of repeat winners without necessarily worsening performance (Coviello, Guglielmo, & Spagnolo, 2018), but it can also be used to favour connected firms, at a measurable cost (Baltrunaite, Giorgiantonio, Mocetti, & Orlando, 2021; Bafanek & Titl, 2024; Decarolis, Fisman, Pinotti, & Vannutelli, 2025; Palguta & Pertold, 2017; Szucs, 2024). Relationships that develop with a buyer’s tenure can sustain repeat awards (Coviello & Gagliarducci, 2017). Relational contracts

can make loyalty to proven suppliers valuable (Board, 2011; Calzolari & Spagnolo, 2009). Past performance legitimately informs supplier choice, although it may deter entry (Butler, Carbone, Conzo, & Spagnolo, 2020; Decarolis, Spagnolo, & Pacini, 2016), and negotiation can be efficient for complex purchases (Bajari, McMillan, & Tadelis, 2009). Persistence is therefore not by itself evidence of harm; its consequences depend on prices and quality (Bandiera, Prat, & Valletti, 2009).

2.2 Procurement as a network

Contract-award records are naturally two-mode. Network studies of procurement have examined corruption risk and favouritism (Fazekas & Wachs, 2020; Wachs, Fazekas, & Kertész, 2021), collusion in co-bidding networks (Reeves-Latour & Morselli, 2017; Wachs & Kertész, 2019), and the structure of buyer-supplier transactions (Popa, 2019). Integrity indicators built from award data include single bidding and the concentration of a buyer's spending on one supplier (Fazekas & Kocsis, 2020; Fazekas, Tóth, Abdou, & Al-Shaibani, 2024; Fazekas, Tóth, & King, 2016). I.G. Fountoukidis et al. (2023) show for the European pacemaker market that concentration-type indices can signal competition while centrality reveals dominant suppliers, and Pliatsidis (2024) measures concentration in Greek procurement with network methods. Recent link-prediction evidence directly examines persistent ties (I. Fountoukidis et al., 2026a). We add constrained random-assignment benchmarks and recurrence-specific comparisons rather than claiming that a move from nodes to ties is novel.

2.3 The Turkish context

Turkish public procurement is governed by Law No. 4734 (2002) and administered by the Public Procurement Authority (KİK), whose platform EKAP publishes notices and awards (Alsaç, 2017); see Buğra and Savaşkan (2014) and Gürakar (2016) on its political economy. Two grounds for negotiation matter here. Under Article 21(b), which covers urgency after unforeseen events, and Article 21(f), which covers purchases of goods and services below an annually indexed limit, the buyer need not publish a notice and instead invites at least three firms of its choosing. Law No. 7144 (2018) extended 21(b) to certain works (Alyanak, 2022). Politically connected firms have been found to win more often under less competitive procedures (Gürakar, 2016; Gürakar & Bircan, 2019), and market capture intensified during the 2010s (Kutlay & Yıldırım, 2026). In the health sector, Decree-Law No. 694 of August 2017 abolished the Public Hospitals Authority and its provincial hospital unions and returned hospital procurement to the Ministry of Health and to individual hospitals.

3 Data

3.1 Collection and scope

We collected award records of IT-related tenders from EKAP by keyword search (13,418 records with an identifiable winner). EKAP search matches substrings of titles, so it admits tenders in which an IT term appears only incidentally. For example, the

keyword *ERP* matches *tuz serpici* (salt spreader), and *technical support* matches hospital staffing. An earlier count-based cleaning left 13,024 contracts, but a value-based audit showed that errors were concentrated among large contracts. The original audit classified 150 high-value contracts by hand using the then-recorded numeric amounts, applied ordered title rules to the rest, and inspected samples from every rule. Correcting currency conversion changes that top-150 ranking by one excluded non-IT record; an additional AI-assisted check against the official tender notice supports its non-IT scope, without verifying its award value or constituting human annotation (Supplementary Information). Of the 13,024 contracts, 1,627 (12.5%) are not IT, such as housing construction, vehicles, and cleaning services. A further 1,406 are ambiguous, such as data-entry staffing or building works with security cameras. The main sample contains the 9,981 unambiguous IT contracts and 10 contracts for IT-enabled call-centre services, one of which is the largest in the data (the 2025 national hospital appointment call centre; the category also contains IT help-desk systems and services); value-based results are reported with and without it. EKAP lists one winner per tender, so multi-lot tenders appear with a single winner (116 titles suggest lots); we found no framework-agreement call-offs.

3.2 Entities, categories, and values

Firm names were canonicalized by normalizing characters and stripping legal-form suffixes, with the merge groups of the 300 largest firms reviewed by hand. This reduces 4,788 recorded names to 4,389 firms, 2,814 in the main sample. Generic buyer labels that pool units across provinces, notably the Ministry of Health’s provincial directorates, were split by the province recorded for each tender (34 labels, 732 main-sample contracts). Each contract is assigned to one of 14 product categories from its title and each buyer to one of 11 sectors.

For an agreement audit, two AI coders based on different language models separately labelled 300 randomly sampled titles, blind to the rules and each other’s outputs. Their agreement was 89% ($\kappa = 0.87$). Rule agreement with each coder was 74.7% and 73.0%; conditional on the 266 titles where the models agreed, it was 79.3%. Class-specific agreement is strongest for health information systems and weaker for custom software and maintenance (Supplementary Information). These comparisons have no human gold standard: model agreement is not accuracy, and conditioning on consensus excludes 34 difficult or disputed cases. Human-referenced evaluation is a separate requirement (Calderon, Reichart, & Dror, 2025); a prospective protocol and existing blind coding sheets are supplied, but completed independent human labels are not available.

Values are first expressed in nominal Turkish lira and then deflated to 2025 TRY using the consumer price index. For 67 main-sample records explicitly denominated in foreign currency, conversion uses the midpoint of the Central Bank of the Republic of Türkiye’s buying and selling rates at the tender date (or the preceding available business day). Currency is unspecified for 845 main-sample records; these are assumed to be TRY and flagged, rather than independently verified. Tender-date conversion is a valuation convention, not an observed settlement rate. Winners that are natural

Table 1 Main sample (IT and IT-enabled services, EKAP 2010–2026).

Awarded contracts	9,991
Firms (canonicalized) / buyers (pooled labels split)	2,814 / 2,890
Buyer–firm pairs; with two or more contracts	7,219; 1,377 (19.1%)
Repeat-eligible contracts (buyer bought in the market before)	4,913
Total value, 2025 TRY (nominal TRY)	89.7 bn (33.3 bn)
Largest contract, share of real value	8.4%
Procedure: open / Art. 21(b) / Art. 21(f) / other	59.0 / 10.5 / 29.6 / 0.9%
Tender dates	Oct 2010 – Mar 2026

persons or lack an explicit legal form are pseudonymized in the released data. Tables 1 and 2 describe the sample.

The extract is not a census of national IT procurement. Keyword searches can omit relevant titles as well as retrieve irrelevant ones, and recall outside the search frame is unknown. Both 2010 and 2026 are partial observation years; history before October 2010 is unavailable.

3.3 Estimated costs

For a blind sample of 150 repeat-eligible tenders, 75 won by incumbents and 75 by other firms, we retrieved the estimated cost (*yaklaşık maliyet*), contract value, and product codes (OKAS, the Turkish version of the EU Common Procurement Vocabulary, CPV) from EKAP’s tender-detail pages. An AI agent read these pages in the authors’ ordinary browser session at a slow pace; its input list did not indicate which tenders were won by incumbents. Collection stopped at 144 tenders when the site displayed a human-verification check, which we did not attempt to bypass. The EKAP contract values match our records within 1% in 99% of cases, and 143 tenders have both values in lira.

4 Methods

4.1 Concentration

For each product category we compute the supplier HHI ($H = \sum_i s_i^2$, 0–10,000) and the four-firm ratio from contract counts and real values. Because single contracts can dominate value shares, we also report value HHIs without the largest contract and with winsorized values, with bias-shifted, independent-contract bootstrap intervals. These quantify resampling variation within the captured records; they do not account for buyer/firm dependence, keyword omission, or classification and entity-resolution errors. As reference points we use 1,000 and 1,800, the “moderately” and “highly” concentrated boundaries of the 1982 and 1992 US merger guidelines; the 2023 guidelines returned to 1,800 for highly concentrated markets (U.S. Department of Justice and Federal Trade Commission, 1992, 2023). These are descriptive reference lines, not Turkish procurement rules or a legal diagnosis; 1,000 is a historical reference, not the 2023 high-concentration threshold.

Table 2 Product categories (main sample). Eligible: repeat-eligible contracts. Value in bn 2025 TRY; procedure shares in % of contracts.

Product category	Contracts	Eligible	Firms	Value	21(b)	21(f)
Health information systems	2,189	1,252	205	19.48	34.7	9.4
ERP / management software	1,515	745	599	11.23	4.5	47.1
Custom software / web / mobile	1,117	488	734	7.07	2.9	43.2
Software licences	1,047	602	375	7.78	1.8	23.4
Network / data-centre infrastructure	748	328	390	7.44	5.9	25.0
Computers / peripherals	696	281	467	2.87	3.9	27.7
Maintenance / support services	626	375	297	8.33	3.5	33.2
GIS / city information	553	230	203	4.75	2.2	41.4
Cybersecurity	390	161	194	3.25	5.4	33.1
Physical security / surveillance	385	182	222	2.29	1.8	23.1
Other IT	277	112	215	1.83	5.4	27.8
Education technology	205	80	128	1.87	5.4	50.2
Smart city / traffic	169	58	106	3.55	4.1	34.3
Call centre / help desk	74	19	46	8.00	5.4	51.4
All	9,991	4,913	2,814	89.74	10.5	29.6

4.2 Incumbency, recurring purchases, and null models

A contract awarded by buyer k in category m on date t is *repeat-eligible* if k awarded another contract in m strictly before t , and *incumbent* if its winner had won from k in m strictly before t . The incumbency rate is the share of repeat-eligible contracts that are incumbent. We flag a *recurring purchase* when the same buyer purchased in the category 6–18 months earlier with title-word Jaccard similarity of at least 0.5 after normalization and stemming. All remaining eligible contracts are *other purchases*; they may include reworded or differently timed recurring services. The flag establishes neither a legal renewal nor whether a need is new. In the AI audit of 60 title pairs, 35–40 of 42 above-threshold pairs and 6–10 of 18 below-threshold pairs were judged the same service, depending on the coder. The direction of bias from classification error is not known.

We permute winner labels within cells and recompute the histories and statistics (1,000 permutations each). Contracts, buyer identities, dates, and each supplier’s win count within a cell remain fixed; exact participation and within-year entry dates are not preserved. Table 3 lists the restrictions. N4 and N4b additionally condition on a prior modal award-province proxy inferred from its prior wins; this is not registered supplier location. The proxy can itself reflect earlier ties with the buyer. Their high null rates partly reflect restrictive and sometimes immovable cells, not proof of exchangeability or universally conservative inference. We use $(1 + b)/(B + 1)$, where b counts draws at least as extreme as the observation. With $B = 1,000$, the minimum is $1/1,001$, reported as $p \leq 0.001$.

4.3 Conditional award associations

The nulls randomize winners among all firms in a cell. A complementary model compares the winner with historically active alternatives; their actual participation and technical eligibility are unobserved. For each repeat-eligible contract we take as alternatives all firms that won in the same market in the 24 months before the tender. We then estimate a conditional logit of who wins among them (McFadden, 1974, 1978). This describes the outcome of participation and award together, not the buyer’s choice alone, and does not identify the buyer’s preference independently of bidding and technical eligibility. The covariates are:

- whether the firm supplied the buyer in the market before (incumbent);
- whether it supplied the buyer in another market;
- whether its prior modal award province matches the buyer’s;
- its recent activity in the market, total experience, and experience with buyers of the same sector.

All covariates are measured before the tender, and standard errors are clustered by buyer. The use of AI tools in coding, agreement checks, and data collection is described in Section 3 and the Declarations. Winners absent from the observed category over the preceding 24 months account for 39% of repeat-eligible contracts. They include returning or previously unobserved firms, not necessarily economic entrants. The main model excludes those contracts; augmenting each set with its observed winner is a sensitivity specification, not a lower bound.

We add a future-tie indicator for a firm that has not previously supplied the buyer in the category but does so at a strictly later date. Future ties are outcomes: winning today can affect later relationships, so the indicator is not an exogenous placebo. Supplementary simulations compare these associations under specified dynamic-history and random-intercept benchmarks, conditional on observed alternative pools and fixed awards. Their calibration and initial conditions are assumptions rather than estimated causal mechanisms (Train, 2009; Wooldridge, 2005).

4.4 Procedure, bids, and prices

We relate incumbency to the procedure with a contract-level logit with product-market, year, and buyer-sector fixed effects and standard errors clustered two-way by buyer and winner (Cameron, Gelbach, & Miller, 2011), and describe its timing with an event study (Supplementary Information). Bid counts come from result announcements for a year-stratified random sample of 393 tenders (2012–2026, design-weighted) and a larger non-random sample of 1,194 software tenders from 2021–2026. In the estimated-cost sample we compare the discount, $1 - \text{contract value/estimated cost}$, between incumbent and other winners by ordinary least squares with heteroskedasticity-robust standard errors.

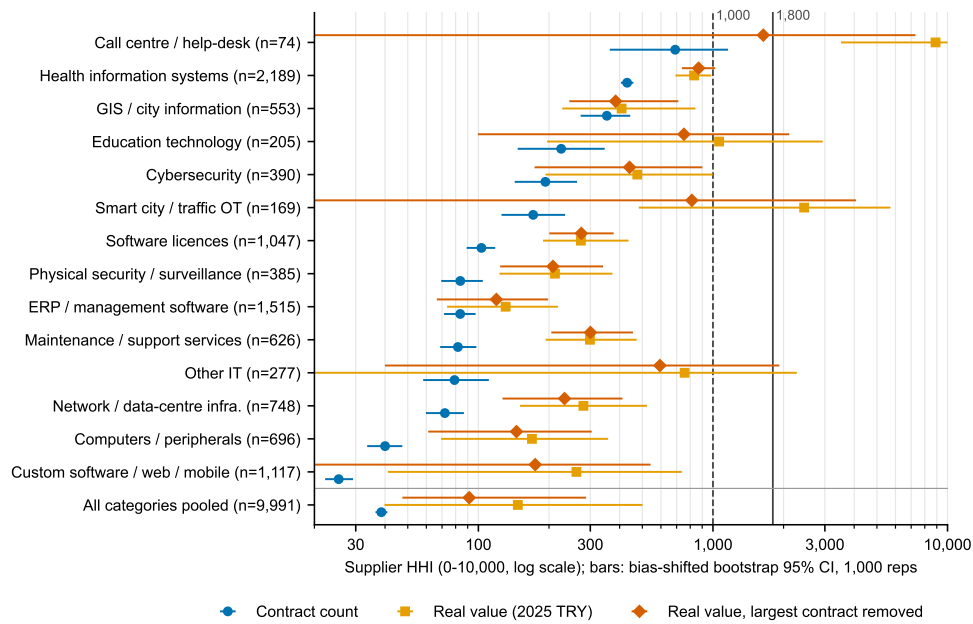


Fig. 1 Supplier concentration by product category: HHI by contract count, by real value, and by real value without the market’s largest contract, with conditional contract-resampling 95% intervals (1,000 bias-shifted resamples). Lines mark 1,000 and 1,800. Log scale.

5 Results

5.1 Award concentration depends on the measure and period

When awards are pooled over the observation period, every category’s count HHI is below 1,000 (Fig. 1); the highest are call-centre services (690, 74 contracts) and health information systems (431, 95% CI 405–458). These figures describe the distribution of captured awards, not a complete relevant market. Value shares and shorter reference periods give a different picture: median annual value HHIs exceed 1,000 in six of the ten categories with qualifying years (at least 30 awards per year), including health information systems, GIS, cybersecurity, and maintenance (Supplementary Information). Thus the dispersed full-period count distribution coexists with substantially higher concentration of expenditure in particular years. Neither measure establishes the alternatives available for an individual tender. The Commission’s market-definition guidance recognizes annual reference periods as a starting point and longer windows for lumpy procurement (European Commission, 2024); our pooled, annual, and sensitivity calculations should be read together.

5.2 Buyers return to incumbents

Of the 4,913 repeat-eligible contracts, 45.2% went to an incumbent. The nulls in Table 3 predict between 6.2% and 26.4%, depending on how much of the market’s

Table 3 Incumbency rate against permutation nulls: winners are permuted within the listed cells (1,000 permutations each; all $p \leq 0.001$). Recurring: a similar title bought by the same buyer in the same market 6–18 months earlier. A dash: not computed for this null.

	Cells: product category $\times$ year $\times$	All	Recurring	Other
Observed		0.452	0.720	0.305
n		4,913	1,743	3,170
N1	–	0.062	0.092	0.045
N2	buyer sector	0.110	0.166	0.078
N3	buyer province	0.207	0.288	0.163
N4	prior award province	0.173	0.270	0.120
N4b	buyer province $\times$ prior award province	0.264	0.391	0.195
Observed / N4b		1.71	1.84	1.57

structure they hold fixed. Observed incumbency exceeds every null ($p \leq 0.001$); under the most restrictive, N4b, it is 1.7 times the null rate, an observed-minus-benchmark gap of about 920 contracts. Under N4b, 38.4% of eligible contracts lie in cells with only one distinct observed winner, so their winner labels cannot move; earlier histories can still change. Restricting label mobility does not establish competitive exchangeability. The excess holds in every product category under N3 (Fig. 2; for call-centre services, with 19 eligible contracts, only at $p = 0.03$) and, under N1, in every year with more than one eligible contract (Supplementary Information). It is robust to recorded instead of split buyer labels and recorded instead of canonicalized firm names. It is also robust to excluding call-centre tenders, multi-lot tenders, natural-person winners, or named-product licence and maintenance recurring purchases, and to including the ambiguous-scope contracts. Among the 540 buyers with at least five contracts, 51% have a main-supplier share above their own N1 null (39% under N2) after applying the Benjamini–Hochberg procedure at nominal 5% (Benjamini & Hochberg, 1995). The network of buyers and suppliers is only moderately modular once degrees are held fixed (Supplementary Information); what stands out is the persistence of individual ties, not segmentation.

The conclusion is insensitive to several specified category and coverage checks. Against category–year–buyer–province nulls, the observed/null ratio is 2.18 in the baseline, 2.35 after merging ERP, custom software, maintenance, and other IT, and 1.76 within health information systems. With no category distinction in either the buyer history or the null, the ratio is 3.77. Retaining the entire observed history but evaluating awards only from 2013 or 2015 gives 2.17 and 2.13. Each comparison uses 1,000 permutations and has $p = 1/1,001$. These exercises address sensitivity inside the retrieved sample, not omitted-title coverage or causal identification (Supplementary Information).

5.3 Recurring and other purchases

A third of repeat-eligible contracts (35.5%) meet the title-based recurrence flag. Their incumbency rate is 72%, and they account for 57–63% of the excess, depending on the

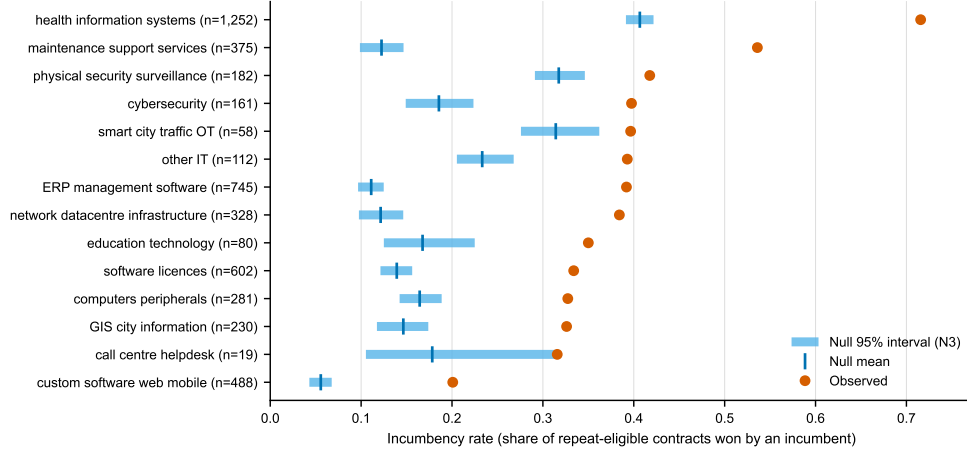


Fig. 2 Incumbency rate by product category: observed (dots) and permutation null N3 (mean and 95% interval; winners permuted within product category $\times$ year $\times$ buyer province). n is the number of repeat-eligible contracts.

null. Other purchases go to incumbents 31% of the time, 1.6 times the most restrictive null (Table 3), and even contracts with no similar earlier title go to incumbents 1.8 times as often as the province null predicts. The difference between recurring purchases and other purchases holds at every interval since the buyer’s previous purchase (Fig. 3) and in each of the eight largest markets. In health information systems, 87% of recurring purchases and 61% of other purchases go to incumbents.

5.4 Conditional associations with past and future ties

Table 4 reports the conditional logit. Its 3,013 contracts have an incumbent share of 68.9%, compared with 45.2% among all repeat-eligible contracts; selection into the recent-winner pool is substantial. Among these 61% of repeat-eligible contracts, a past buyer–supplier tie is associated with 42 times the odds of winning (95% CI 36–49), holding constant its recent activity, experience, and location. Augmenting choice sets with previously inactive or unobserved winners gives an odds ratio of 22, a sensitivity result rather than a bound. The advantage is largest for recurring purchases (141) and under Article 21(b) (96), and smaller for other purchases (21) and open tenders (34).

Future ties also have a large conditional association with current wins: an odds ratio of 54 (42–70), alongside 64 for past suppliers in the model including both indicators. This differs from the 42 in Table 4, which excludes the future indicator. Among other purchases, the future-tie odds ratio is about 40 and the past-tie odds ratio about 30; among title-flagged recurring purchases, the past/future ratio is about two. Equal 730-day observation windows retain this qualitative contrast (Supplementary Information). Inferred alternatives are not observed bidders, and future ties are outcomes. These estimates therefore combine participation, persistent match characteristics, and history-dependent processes. The supplementary simulations examine the sensitivity of lead-lag comparisons to initialization and calibration; they do not allocate

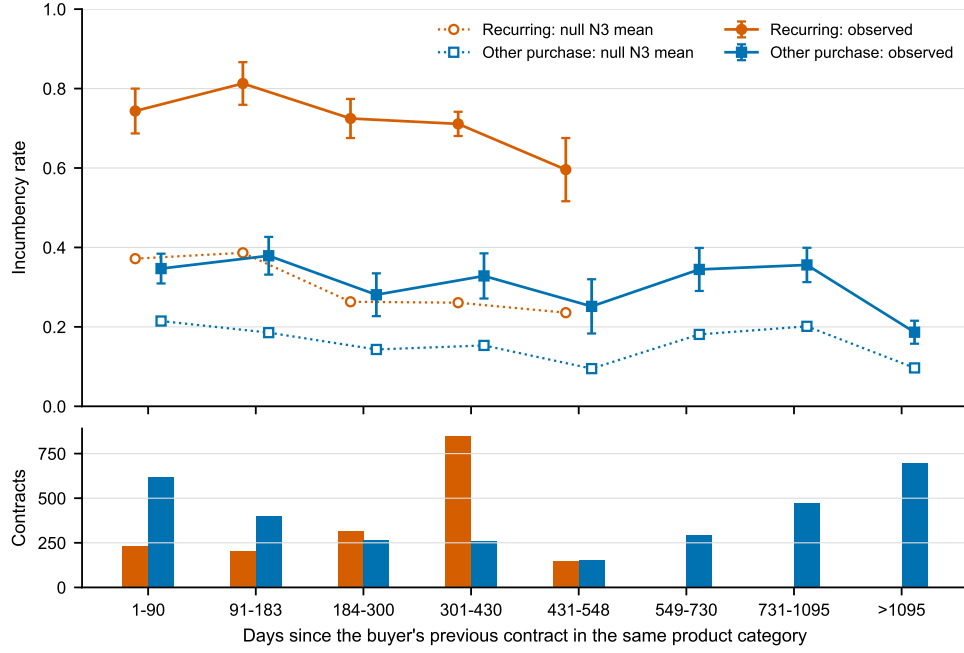


Fig. 3 Incumbency by days since the buyer's previous contract in the same product category, for recurring purchases and other purchases: observed (95% intervals) and null N3 mean. Lower panel: number of contracts.

persistence across mechanisms. With all historical controls recomputed, two shifted-intercept initializations both match the observed incumbent share but produce mean past/future ratios of 1.60 and 2.06. An unshifted version fails to match that share on the tested grid. The diagnostic contrast therefore depends on the initialization assumption (Supplementary Information).

5.5 Procedure

Incumbency is higher under Article 21(b) than under open tendering (72% against 44% of repeat-eligible contracts, a gap well above its null). Most excess incumbency nonetheless occurs in open tenders, which are far more numerous. Under Article 21(f), 35% of repeat-eligible contracts go to an incumbent, 2.6 times the province null. The 21(b)–open gap rose sharply around 2017–18. The rise is concentrated among health buyers and coincides with the decentralization of hospital procurement, although the difference from other buyers is not statistically significant. The exact timing and cause are uncertain, and the Supplementary Information reports the event study and its caveats.

5.6 Bids and prices

Incumbent wins coincide with weaker competition at the tender stage. In the random bid-count sample, tenders won by an incumbent drew a single bid in 71% of cases

Table 4 Conditional logit of who wins among firms active in the market in the 24 months before each tender. Odds ratios with buyer-clustered 95% confidence intervals; 3,013 contracts won by firms active in the market in the previous 24 months (909 buyers). The remaining 1,900 contracts had winners absent from the preceding 24-month category history.

	Odds ratio [95% CI]
Incumbent: supplied this buyer in this market before	42.3 [36.4, 49.2]
Supplied this buyer in another market	2.17 [1.68, 2.81]
Prior modal award province = buyer’s province	2.85 [2.44, 3.33]
Log wins in this market, previous 12 months	1.89 [1.71, 2.10]
Log total prior wins	0.91 [0.83, 1.01]
Log prior wins with buyers of the same sector	1.79 [1.60, 2.00]
Incumbent OR: recurring purchases / other purchases	140.7 / 21.0
Incumbent OR: Article 21(b) / open	95.8 / 33.7
Incumbent OR, observed winner added	21.5 [18.5, 25.0]

(95% CI 61–80%), against 32% (24–42%) for other winners (Fig. 4). An unweighted logit with a negotiated-procedure indicator and year trend gives an odds ratio of 4.4; an unweighted specification distinguishing three procedure groups and controlling for recurrence gives 3.6. In the larger software sample the odds ratio for a single valid bid is 2.0, and 2.7 with a recurrence control. These are associations: incumbents may win because others do not bid, or others may not bid because the incumbent is expected to win.

In the estimated-cost sample, contracts won by incumbents were awarded at a mean discount of 12.1% below the estimated cost, against 16.5% for other winners (medians 8.9% and 11.2%). The difference is 4–6 percentage points with or without controls for recurrence, procedure, value, year, and market. It is not statistically significant (p between 0.11 and 0.22, $n = 143$). The estimated cost is an administrative benchmark, not a measure of marginal cost; these imprecise discount associations establish neither a markup nor a switching-cost effect.

6 Discussion

6.1 Interpretation

Aggregate award shares and buyer histories provide complementary information. Dispersed full-period counts coexist with higher annual value concentration and a substantial excess of repeat awards relative to constrained assignment nulls. Both past and future ties are associated with current wins. This makes a simple causal reading of the incumbent coefficient inappropriate, but it does not quantify persistent heterogeneity or show that switching costs occur only in recurring purchases. Specialization, compatibility, satisfactory performance, anticipated project stages, learning, and preferential relationships remain observationally intertwined (Coviello et al., 2018; Decarolis et al., 2016; Gürakar, 2016; Kutlay & Yıldırım, 2026).

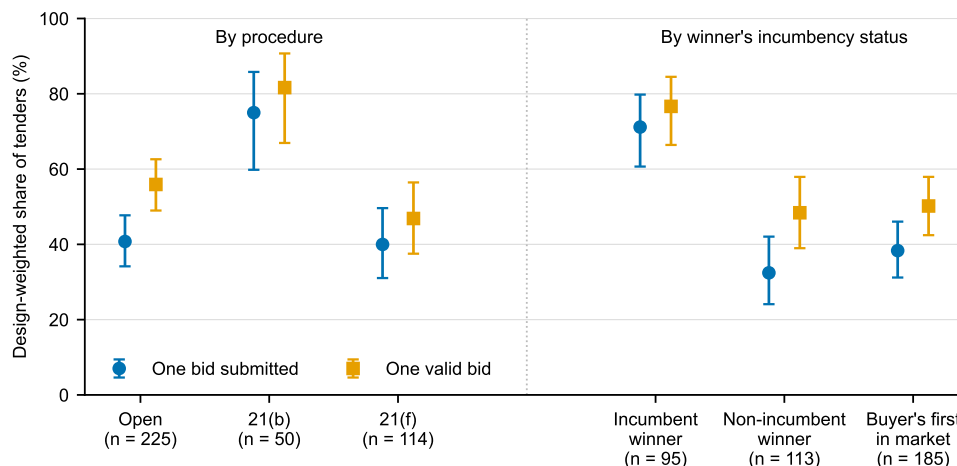


Fig. 4 Single-bid rates by procedure and by whether the winner was an incumbent (design-weighted, year-stratified sample of 393 tenders, 2012–2026; 95% intervals).

Single bids are much more frequent among incumbent-won tenders. In a small sample, they also carry smaller discounts on the estimated cost. Both patterns fit weaker competition when incumbents are present. Both also fit purchases that only the incumbent can supply. Procedure is associated with persistence but its causal role is not identified: persistence is highest under Article 21(b), but most of it arises in open tenders.

6.2 Implications for monitoring

Buyer–supplier concentration and relationship-based screens already appear in the integrity and predictive literatures (Fazekas & Kocsis, 2020; Fazekas et al., 2024; I. Fountoukidis et al., 2026a). Our contribution is to make the assignment benchmark and recurrence definition explicit and to show how their conclusions change under observable classification and history restrictions. The comparison changes the descriptive magnitude: the benchmark incumbent rate rises from 6.2% with category–year supplier totals held fixed to 26.4% with both province proxies added. The observed-minus-benchmark gap consequently falls from about 39 to 19 percentage points. Raw persistence alone obscures how much the chosen assignment restrictions matter; the remaining gap is not an estimate of inefficient or preferential procurement. High persistence identifies records for contextual review, not violations or a particular causal mechanism. Case-level analysis would need bidder identities, technical requirements, installed-system compatibility, prices, and performance. Interoperability and data portability may address demonstrated switching barriers (European Commission, 2013), but these award patterns alone do not establish that such a remedy is appropriate for a given buyer.

6.3 Limitations

The evidence is observational and conditional on keyword capture. We lack a national sampling denominator, prior history before October 2010, losing bidders' identities, technical quality, and reliable independent classification labels. Titles proxy product categories and repeated services, not relevant markets or contractual renewals. The AI audit does not substitute for human validation; independent coding and reconciliation under the supplied prospective protocol remain outstanding. Conditional-logit alternatives are inferred from past winners and exclude many observed awards. Future-tie variables condition on later outcomes, while simulations retain observed histories and use calibration and initialization choices that preclude causal decomposition. Estimated costs for 143 tenders do not establish welfare effects. Firm-name canonicalization lacks tax identifiers, and multi-lot tenders appear with a single winner. Reverting to recorded firm names or buyer labels changes observed incumbency to 0.394 or 0.462, respectively, although excess persistence remains.

7 Conclusion

In the retrieved Turkish public IT awards, buyers repeatedly choose previous suppliers more often than constrained random assignment predicts. This excess persists under coarser categories, category-free histories, and later evaluation windows, and is especially large for title-defined recurring purchases. Concentration depends on value weighting and the time interval; low pooled count concentration does not establish competitive relevant markets. Lead-lag estimates and simulations illuminate the assumptions needed to interpret persistence but do not identify switching costs or distinguish benign from preferential relationships. The defensible application is complementary monitoring of award concentration and buyer-level continuity, followed by case-specific investigation.

Supplementary information. The Supplementary Information contains the scope audit and classification checks, concentration tables, all null-model results and robustness checks, the recurrence classification, the choice model and simulation details, the procedure and timing analysis with the statutory texts, a dyad-continuation model, the bid and price samples, and the network analysis.

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Declarations

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- **Competing interests.** The authors have no competing interests to declare.
- **Ethics approval.** Not applicable. The study uses public administrative records; names of natural persons are pseudonymized in the released data.

- **Consent to participate.** Not applicable.
- **Consent for publication.** Not applicable.
- **Data and code availability.** The revised code, derived data, and data dictionary are provided in TenderNet v1.1.0 (<https://github.com/FurkanAksoyy/TenderNet/releases/tag/v1.1.0>). The version-spanning Zenodo DOI is <https://doi.org/10.5281/zenodo.22994070>; select version 1.1.0 for the replication package accompanying this manuscript. The underlying records are published by the Public Procurement Authority on EKAP (<https://ekapv2.kik.gov.tr>).
- **Use of AI tools.** Claude (Anthropic) assisted with analysis code, figure preparation, language editing, and data collection. Codex (OpenAI) assisted with the revision, literature checking, code audits, and robustness analysis. For the classification agreement audit, two instances based on different models separately labelled samples of tender titles (disclosed in Section 3). An AI agent also retrieved estimated costs from EKAP’s public pages through the authors’ own browser at a slow pace, without bypassing any verification. The authors designed the study, checked the code and results, and take full responsibility for the content.
- **Author contributions.** F.A.: conceptualization, methodology, data curation, software, formal analysis, visualization, writing (original draft, review and editing). A.Ş.: data curation, investigation, validation, visualization, writing (review and editing).

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